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A Computational Turn? Reflections on Digital and Computational History of Science, Knowledge and Technology

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With digital and computational methods becoming more visible both in scholarly and everyday contexts, it is timely to ask whether a distinct field of digital and computational history of science, knowledge, and technology can meaningfully be said to exist. This epilogue to the special issue on computational approaches to the history of alchemy and chemistry addresses this question, arguing that such an overarching field has not yet materialised. To fulfil its programmatic promises, a digital and computational history of science, knowledge, and technology must focus on conceptual clarity regarding its disciplinary position. If digital and computational approaches are to contribute meaningfully to the history of science and knowledge, they should support efforts to produce more global and inclusive historiographies rather than reproducing existing biases in available datasets and algorithms. Indeed, one of the most important tasks for the future may be the creation of datasets for under-resourced subfields and historical questions. Thus, this paper argues for a commitment to diversity in the subjects, cultures, languages, periods, and forms of knowledge considered: Only through employing a plurality of methods can a computational history of science, knowledge, and technology adequately reflect the diversity of knowledges represented within the broader field.

Introduction

This special issue on “Computational Approaches to History of Alchemy and Chemistry” invites reflection on a broader question: what would a digital and computational history of science and knowledge look like as an established field of

research? A central premise of this epilogue is that, while several influential publications have invoked a “computational history of science,” an established field of that name has yet to emerge, even less so for a computational history of chemistry.¹ Important projects, experienced practitioners, and a growing body of scholarship certainly exist, but they have not yet coalesced into a field with shared scholarly publication venues, conferences, infrastructures, institutions, and a clearly defined community of practice.²

This observation also invites closer scrutiny of the terminology itself. Earlier interventions largely adopted the language of computational history of science.³

¹ Manfred D. Laubichler, Jane Maienschein, and Jürgen Renn, “Computational Perspectives in the History of Science: To the Memory of Peter Damerow,” *Isis* 104, no. 1 (2013): 119–30; Abraham Gibson, Manfred D. Laubichler, and Jane Maienschein, “Introduction (Focus: Computational History and Philosophy of Science),” *Isis* 110, no. 3 (2019): 497–501. Cf. Arianna Betti and Hein van den Berg, “Towards a Computational History of Ideas,” in *Proceedings of the Third Conference on Digital Humanities in Luxembourg with a Special Focus on Reading Historical Sources in the Digital Age: Luxembourg, Luxembourg, December 5–6, 2013*, ed. Lars Wieneke, Catherine Jones, Marten Düring, Florina Armaselu, and René Leboutte, *CEUR Workshop Proceedings* 1681 (Aachen: CEUR-WS, 2016), <https://ceur-ws.org/Vol-1681/>, http://ceur-ws.org/Vol-1681/Betti_van_den_Berg_computational_history_of_ideas.pdf (accessed 3 May 2026); Giulia Giannini, “Towards a Computational History of Science: Limitations and Perspectives of an Emerging Research Approach,” *Physis: International Journal for the History of Science* 57, no. 1 (2022): 245–58; Jürgen Jost et al., “Computational History: Challenges and Opportunities of Formal Approaches,” *Journal of Social Computing* 4, no. 3 (2023): 232–42; Sarah Lang, Bernhard Liebl and Manuel Burghardt, “Toward a Computational Historiography of Alchemy: Challenges and Obstacles of Object Detection for Historical Illustrations of Mining, Metallurgy and Distillation in 16th–17th Century Print,” *Computational Humanities Research* (2023): 29–48, <https://ceur-ws.org/Vol-3558/paper342.pdf>.

² Some examples are: Rob Iliffe, ed., *The Newton Project*, <https://www.newtonproject.ox.ac.uk/> (accessed 3 May 2026); William R. Newman, ed., *The Chymistry of Isaac Newton* (Bloomington: Indiana University, 2005–2026), <http://chymistry.org> (accessed 3 May 2026); On the Newyton Chymistry project: Cesare Pastorino, Tamara Lopez, and John A. Walsh, “The Digital Index Chemicus: Toward a Digital Tool for Studying Isaac Newton’s *Index Chemicus*,” *Body, Space & Technology* 7, no. 2 (2008), <https://doi.org/10.16995/bst.141>; John A. Walsh and Wallace Edd Hooper, “The Liberty of Invention: Alchemical Discourse and Information Technology Standardization,” *Literary and Linguistic Computing* 27, no. 1 (2012): 55–79, <https://doi.org/10.1093/lle/fqr038>; Levi King, Sandra Kübler, and Wallace Edd Hooper, “Word-Level Language Identification in *The Chymistry of Isaac Newton*,” *Digital Scholarship in the Humanities* 30, no. 4 (2015): 532–40, <https://doi.org/10.1093/dls/duq032>; Warren Christopher N., Daniel Shore, Jessica Otis, Lawrence Wang, Mike Finegold, and Cosma Shalizi, “Six Degrees of Francis Bacon: A Statistical Method for Reconstructing Large Historical Social Networks,” *Digital Humanities Quarterly* 10, no. 3 (2016); Lauren T. Kassell, “Paper Technologies, Digital Technologies: Working with Early Modern Medical Records,” in *Medical Records and the Life Cycle of Medical Knowledge*, ed. Lauren Kassell et al. (Edinburgh: Edinburgh University Press, 2016), 120–35; Marieke M. A. Hendriksen and Ruben E. Verwaal, “Boerhaave’s Furnace: Exploring Early Modern Chemistry through Working Models,” *Berichte zur Wissenschaftsgeschichte* 43 (2020): 385–411; Tara Nummedal and Donna Bilak, eds., *Furnace and Fugue: A Digital Edition of Michael Maier’s Atalanta fugiens (1618) with Scholarly Commentary* (Charlottesville: University of Virginia Press, 2020), <https://furnaceandfugue.org/> (accessed 3 May 2026), contextualised in: Sarah Lang, “Furnace and Fugue: A Multimedia Edition of Michael Maier’s *Atalanta fugiens*,” *RIDE: A Review Journal for Scholarly Digital Editions and Resources* 18 (2023), <https://doi.org/10.18716/ride.a.18.4>. Pamela H. Smith, et al., eds., *The Making and Knowing Project: Intersections of Craft Making and Scientific Knowledge* (2020), <https://www.makingandknowing.org/bnf-ms-fr-640/> (accessed 3 May 2026), contextualised in: Sarah Lang, “Ways of Making Digital Editions of Historical Recipes and Experimental Texts: The Making and Knowing Project,” *RIDE: A Review Journal for Scholarly Digital Editions and Resources* 21 (2026), <https://ride.i-d-e.de/issues/issue-21/makingandknowing/>; Matteo Valleriani, *The Sphere: Knowledge System Evolution and the Shared Scientific Identity in Europe* (Berlin: Max Planck Institute for the History of Science, 2015–2026), <https://www.mpiwg-berlin.mpg.de/project/the-sphere> (accessed 15 June 2026).

³ For instance, many of those cited in the previous footnote. The co-editors of this special issue have followed in the footsteps of these earlier publications and equally chosen this name for the special issue, which is contextualised more in the introduction: Farzad Mahootian, Sarah Lang and Guillermo Restrepo, “Introduction: Computational Approaches to the Histories of Alchemy and Chemistry,” *Ambix*, this issue.

Yet developments within the umbrella discipline of digital humanities (DH) increasingly distinguish between digital humanities as a broad domain and computational humanities, to which computational history of science (and chemistry) belongs as one of its subfields.⁴ From this perspective, the more pressing question may be why discussions have focused almost exclusively on a computational history of science without simultaneously articulating a digital history of science and knowledge?⁵

The distinction is not merely terminological. Computational approaches depend upon datasets, infrastructures, and forms of scholarly labour that are typically associated with digital humanities more broadly. Digitisation, transcription, scholarly editing, metadata creation, annotation, and corpus construction constitute the foundations upon which computational analysis depends. A computational history of science and knowledge can therefore only develop on the basis of a broader digital history of science and knowledge. If the former has remained relatively underdeveloped, one may argue that this is because the latter has never fully consolidated as a recognised field in its own right. Computational Literary Studies, which is a prime example of a DH subdiscipline that has become established as a field in its own right, for example, benefits from decades of investment in digital editing, corpus construction, text encoding, and the creation of large-scale textual resources.⁶ Historians of science and knowledge likewise possess numerous valuable datasets, digital editions, databases, and digital projects, but these resources remain dispersed and have rarely been conceptualised as components of a shared digital history of science and knowledge.⁷ The challenge, therefore, is not simply to establish a computational history of science and knowledge, nor to explore the computational history of alchemy and chemistry with that aim, but rather to foster a broader digital and computational history of science, knowledge, and technology.⁸ This inclusive formulation reflects the relationship between infrastructure

⁴ David M. Berry, "The Computational Turn: Thinking about the Digital Humanities," *Culture Machine* 12 (2011): 1–22. Camille Roth, "Digital, Digitized, and Numerical Humanities," *Digital Scholarship in the Humanities* 34, no. 3 (2019): 616–32, <https://doi.org/10.1093/lc/fqy057>. Lauren Tilton, David Mimno, and Jessica Marie Johnson, eds., *Computational Humanities, Debates in the Digital Humanities* 11 (Minneapolis: University of Minnesota Press, 2024), <https://doi.org/10.5749/9781452973098>; Sarah Lang, "(Doing) Computational History: On the Role of Data Work in Computational Approaches," *Histories* 6, no. 2 (2026): 26, <https://doi.org/10.3390/histories6020026>.

⁵ After all, there is already an established field operating under the name of *digital history*. See Daniel J. Cohen and Roy Rosenzweig, *Digital History: A Guide to Gathering, Preserving, and Presenting the Past on the Web* (Philadelphia: University of Pennsylvania Press, 2005), <https://chnm.gmu.edu/digitalhistory/> (accessed 8 June 2026). Simone Lässig, "Digital History: Challenges and Opportunities for the Profession," *Geschichte und Gesellschaft* 47 (2021): 5–34.

⁶ Jan Luhmann and Manuel Burghardt, "Digital Humanities—A Discipline in Its Own Right? An Analysis of the Role and Position of Digital Humanities in the Academic Landscape," *Journal of the Association for Information Science and Technology* 73, no. 2 (2022): 148–71.

⁷ Past contributions reflecting on future trends in the historiography of alchemy have likewise suggested that digital methods are likely to become an increasingly important area of research and methodological development: Marcos Martínón-Torres, "Some Recent Developments in the Historiography of Alchemy," *Ambix* 58, no. 3 (2011): 215–37; Sarah Lang, "Alchemical Laboratories: Texts, Practices, Material Relics: An Introduction," in *Alchemische Labore: Texte, Praktiken und materielle Hinterlassenschaften / Alchemical Laboratories: Texts, Practices, Material Relics*, ed. Sarah Lang (Graz: Graz University Library Publishing, 2023), 15–40, <https://doi.org/10.25364/97839033740412>.

and analysis, data creation and data use, and digital humanities and computational humanities. Computational analysis depends upon digital infrastructures, while digital scholarship increasingly benefits from computational tools.⁹ Neither can flourish independently of the other. In the end, developing what is currently a scholarly trend into a field in its own right thus requires clarity about what is meant by both *digital* and *computational* approaches.

At the same time, it is necessary to remain attentive to the ways in which computational research is shaped by the availability of datasets and algorithms.¹⁰ Amongst the materials already available, the most amenable to computational analysis are not necessarily representative of the historical record, whilst existing infrastructures and tools often reflect biases towards modern, Western, and especially English-language sources.¹¹ I therefore argue that the challenge is not simply to apply new methods, but to build a field that is sufficiently open and pluralistic to accommodate different forms of expertise, sources, and traditions of knowledge. Doing so requires recognising the diversity of the sources, media, languages, regions, and periods that constitute the history of science and knowledge. Indeed, a field broad enough to accommodate this diversity cannot be built around a single method or technological paradigm. Instead, it should remain pluralistic, drawing on a wide range of digital and computational approaches while remaining attentive to the ways in which datasets, infrastructures, and algorithms shape what

⁸ If a more substantial computational history of science and knowledge is to emerge, one of the central tasks will be to connect these dispersed efforts more effectively. This requires both methodological innovation and community building. Historians of science are well positioned to reflect critically on technologies, algorithms, and the production of knowledge, yet meaningful engagement with digital methods requires some familiarity with their inner workings, as well as their concepts, histories, and limitations. The issue is not that every historian of science should become a computational specialist, but that a sufficient level of digital literacy is needed to facilitate informed dialogue between specialists and non-specialists. Such dialogue could benefit both sides. Digital humanities venues often provide audiences with strong methodological expertise but less familiarity with particular historical subjects, whereas historians of science possess precisely that contextual knowledge. Bringing these conversations into closer contact could arguably strengthen the field as a whole. The aim is not to replace established historical approaches, but to place them in productive dialogue with digital and computational methods.

⁹ See Lang, "(Doing) Computational History"; For some digital enrichment methods, see Sarah Lang, "Digitale Erschließungsmethoden für alchemische Texte am Beispiel der *Symbola Aureae Mensae* Michael Maiers," in *Alchemie – Genealogie und Terminologie: Bilder, Techniken und Artefakte*, ed. Petra Feuerstein-Herz and Ute Frietsch, Wolfenbütteler Forschungen 166 (Göttingen: Harrassowitz, 2022), 133–52; Sarah Lang, "A Machine Reasoning Algorithm for the Digital Analysis of Alchemical Language and Its *Decknamen*," *Ambix* 69, no. 1 (2022): 65–83, <https://doi.org/10.1080/00026980.2022.2038428>; Sarah Lang, "Vom 'Wissen in Buchform' zum formalen Wissensmodell: Digitale Aufbereitung wissenschaftshistorischer Drucke am Beispiel der *Alchemica* Michael Maiers," in *Wissen und Buchgestalt*, ed. Philipp Hegel and Michael Krewet (Wiesbaden: Harrassowitz, 2022), 227–248.

¹⁰ See Lang, "(Doing) Computational History."

¹¹ See for example Amy E. Earhart, "Digital Humanities within a Global Context: Creating Borderlands of Localized Expression," *Fudan Journal of the Humanities and Social Sciences* 11 (2018): 357–69, <https://doi.org/10.1007/s40647-018-0224-0>; Roopika Risam, *New Digital Worlds: Postcolonial Digital Humanities in Theory, Praxis, and Pedagogy* (Evanston, IL: Northwestern University Press, 2018); Domenico Fiormonte, Sukanta Chaudhuri, and Paola Ricaurte, eds., *Global Debates in the Digital Humanities* (Minneapolis: University of Minnesota Press, 2022), <https://www.upress.umn.edu/book-division/books/global-debates-in-the-digital-humanities> (accessed 8 June 2023); On the resource gap, see Sarah Lang et al., "Confabulated Transcriptions?" Facet Publishing London; Sarah Lang and Elena Suárez Cronauer, *Beyond Data Feminism: Towards Ethical Data Work in the (Digital) Humanities* (= *Zeitschrift für digitale Geisteswissenschaften / Working Papers*, 4) (Herzog August Bibliothek: Wolfenbüttel, 2026), https://zfdg.de/wp_2026, https://doi.org/10.17175/wp_2026 (accessed 3 May 2026); Sarah Lang and Elena Suárez Cronauer, "Dataset Audits for Mitigating Data Gaps," *Computational Humanities Research* (accepted, forthcoming 2026).

can be studied in the first place.¹² Arguably, it should also draw upon a wide range of digital and computational approaches, allowing different research questions to be addressed with methods suited to their particular contexts and affordances. Thus, a collective effort is needed to fill the outline sketched by earlier programmatic discussions and to articulate what a computational turn in the history of science and knowledge might mean in practice.

Situating the field of digital and computational history of science, knowledge and technology, and under this umbrella, of chemistry in particular, in this way also highlights the importance of remaining connected both to the wider digital humanities community and to the broader history of science, knowledge, and technology community.¹³ Many of the challenges currently confronting computational approaches, such as issues with datasets, modelling, bias, sustainability, documentation, and openness, have already been debated extensively within digital humanities.¹⁴ Therefore, there is much more to be gained from participating in these discussions than from reproducing them within a separate disciplinary space.

However, the present situation offers an opportunity. If a digital and computational history of science and knowledge remains in the process of taking shape, there is still considerable scope to define what such a field should encompass. The following sections discuss aspects that should be taken into account in the process.

Disciplinary trajectories and earlier programmatic interventions

To assess where a digital and computational history of science and knowledge currently stands, it is useful to situate it within the broader development of digital

¹² Domenico Fiormonte, "Towards a Cultural Critique of the Digital Humanities," *Historical Social Research* 37, no. 3 (2012): 59–76, <https://doi.org/10.12759/hsr.37.2012.3.59-76>; Andrew Iliadis and Federica Russo, "Critical Data Studies: An Introduction," *Big Data & Society* 3 (2016), <https://doi.org/10.1177/2053951716674238>; Roopika Risam, "Beyond the Margins: Intersectionality and the Digital Humanities," *Digital Humanities Quarterly* 9, no. 2 (2015), <http://www.digitalhumanities.org/dhq/vol/9/2/000208/000208.html> (accessed 8 June 2026); Michelle Caswell, Ricardo Punzalan, and T-Kay Sangwand, "Critical Archival Studies: A Special Issue," *Journal of Critical Library and Information Studies* 1, no. 2 (2017): 1–8; Itza A. Carbajal and Michelle Caswell, "Critical Digital Archives: A Review from Archival Studies," *American Historical Review* 126, no. 3 (2021): 1102–20, <https://doi.org/10.1093/ahr/rhab359>; David M. Berry, "Critical Digital Humanities," in *The Bloomsbury Handbook to the Digital Humanities*, ed. James O'Sullivan (London: Bloomsbury Academic, 2022), 125–36, <https://doi.org/10.5040/9781350232143.ch-12>; Domenico Fiormonte and Gimena del Rio Riande, "The Peripheries and Epistemic Margins of Digital Humanities," in *The Bloomsbury Handbook to the Digital Humanities*, ed. James O'Sullivan (London: Bloomsbury Academic, 2022), 19–28, <https://doi.org/10.5040/9781350232143.ch-2>; Roopika Risam, "Decolonizing the Digital Humanities in Theory and Practice," in *The Routledge Companion to Media Studies and Digital Humanities* (Abingdon: Routledge, 2018), 78–86; Roopika Risam, "Postcolonial Digital Humanities Reconsidered," in *The Bloomsbury Handbook to the Digital Humanities*, ed. James O'Sullivan (London: Bloomsbury Academic, 2022), 41–48, <https://doi.org/10.5040/9781350232143.ch-4>; Hannah Smyth, Julianne Nyhan, and Andrew Flinn, "Feminist Digital Humanities and Critical Heritage Studies," in *Routledge International Handbook of Research Methods in Digital Humanities*, ed. Kristen Schuster and Stuart Dunn (Abingdon: Routledge, 2020), 295–308.

¹³ Without such engagement, the field risks becoming doubly isolated: detached from mainstream history of science on the one hand and from the wider digital humanities conversations on the other.

¹⁴ On dataset documentation practices and auditing datasets, see Timnit Gebru, et al., "Datasheets for Datasets," *Communications of the ACM* 64 (2021): 86–92; Lang, "(Doing) Computational History"; Sarah Lang, "Documenting Datasets as a Tool for Change," in *Digital Humanities 2025: Book of Abstracts*, ed. Gimena del Rio Riande, et al. (Lisbon, 2025), 885–87, <https://zenodo.org/records/18393402> (accessed 3 May 2026). Lang and Suárez Cronauer, "Dataset Audits for Mitigating Data Gaps."

humanities and its discipline-specific trajectories. Digital humanities have always been interdisciplinary, but their adoption has varied considerably across fields. English studies, for example, integrated digital methods relatively early, and computational literary studies has since become highly institutionalised, with its own conferences, journals, professorships, and degree programmes.¹⁵ Other fields such as archaeology, linguistics, and, more recently, art history have also developed comparatively strong digital or computational traditions, although each has followed a different path.¹⁶ For the history of science, knowledge and technology, computational approaches have developed more haphazardly.¹⁷ What exists are practitioners, projects, and pockets of expertise rather than a consolidated community of practice. Many scholars working with digital and computational methods have therefore found stronger intellectual homes in digital humanities venues than within the mainstream structures of the history of science itself. This has allowed important work to continue, but it has also left expertise dispersed across different communities.

The earlier contributions heralding a “computational history of science,” despite the quality of the individual articles and the significance of the research questions they addressed, have remained largely programmatic in their attempts to articulate what a broader field might look like if it were fully consolidated. Rather than documenting an already established subdiscipline, these interventions primarily outlined visions, possibilities, and agendas for future development but have also done so largely from the perspective of their own areas of specialisation.

The 2013 *Isis* contribution on the emerging “computational turn,” for example, offered an ambitious vision, anticipating developments such as large-scale analysis, the integration of diverse sources, and computational methods opening new avenues of inquiry.¹⁸ Yet, one may argue that many of these possibilities were still aspirational. The seamless integration of heterogeneous historical sources, for instance, remains dependent on substantial infrastructural and curatorial

¹⁵ Matthew G. Kirschenbaum, “What Is Digital Humanities and What’s It Doing in English Departments?,” in *Defining Digital Humanities: A Reader*, ed. Melissa M. Terras, Julianne Nyhan, and Edward Vanhoutte (London: Routledge, 2013; repr. 2016), 195–203.

¹⁶ On digital art history see: Johanna Drucker, “Is There a ‘Digital’ Art History?,” *Visual Resources* 29, nos. 1–2 (2013): 5–13; Murtha Baca, Anne Helmreich, and Melissa Gill, “Digital Art History,” in *Visual Resources* 35, nos. 1–2 (2019): 1–5. On digital archaeology see: Ezra Zubrow, “Digital Archaeology: A Historical Context,” in *Digital Archaeology: Bridging Method and Theory*, ed. Thomas L. Evans and Patrick T. Daly (London: Routledge, 2006), 10–31; Colleen Morgan, “Current Digital Archaeology,” *Annual Review of Anthropology* 51, no. 1 (2022): 213–31. On computational literary studies (CLS) see: J. Berenike Herrmann, et al., “Tool Criticism in Practice: On Methods, Tools and Aims of Computational Literary Studies,” *Digital Humanities Quarterly* 17, no. 3 (2023); Benjamin Krautter, “The Scales of (Computational) Literary Studies: Martin Mueller’s Concept of Scalable Reading in Theory and Practice,” in *Zoomland: Exploring Scale in Digital History and Humanities* (Bielefeld: transcript, 2024), 261–86.

¹⁷ See for examples the digital projects, data bases, digital scholarly editions and computational studies referenced above in n. 2.

¹⁸ Laubichler, Maienschein, and Renn, “Computational Perspectives in the History of Science.” This moment was certainly inspired by the big data paradigm discussed in society at large and also by authors like David Berry announcing a computational turn in DH. David M. Berry, “The Computational Turn: Thinking about the Digital Humanities,” *Culture Machine* 12 (2011): 1–22. On big data see Elena Aronova, Christine von Oertzen, and David Sepkoski, “Introduction: Historicizing Big Data,” *Osiris* 32 (2017): 1–17.

labour. Its optimism also reflected a wider enthusiasm for scale and *longue durée* analysis, sometimes approaching “end of theory” assumptions that large datasets might overcome the need for interpretation.¹⁹ Subsequent work in critical data studies has shown why this cannot be assumed: digitisation reproduces archival, linguistic, and geographical biases rather than overcoming them.²⁰

In 2019, *Isis* featured a focus section on “Computational history and philosophy of science” attentive to then ongoing developments surrounding Artificial Intelligence (AI) in society at large and current issues in digital history and digital humanities more broadly, situating computational methods within broader historiographical and methodological debates.²¹ At the same time, both the 2013 and 2019 discussions in the journal remained concentrated in particular empirical domains, from the life sciences, like evolutionary biology and medicine.²² Although these examples are valuable, they do not represent the full diversity of science and knowledge traditions. In this sense, earlier claims about a computational history of science rested on relatively narrow evidential foundations that did not yet go beyond modern and Western sources and innovation-focused paradigms of what it means to create knowledge.

A recurring issue in these discussions is the relationship between computation and interpretation. Computational methods are sometimes criticised for merely confirming what historians already knew; when they produce unexpected results, they are sometimes dismissed as unreliable.²³ Both responses are inadequate. As also Guillermo Restrepo argues in his contribution to this special issue, computational methods can reveal patterns, generate evidence, and support interpretation, but they cannot determine historical meaning on their own.²⁴ Algorithms may detect regularities, clusters, or correlations, but their significance only emerges through contextualisation and domain expertise.

¹⁹ The article that started the “end of theory” debate is: Chris Anderson, “The End of Theory: The Data Deluge Makes the Scientific Method Obsolete,” *Wired*, June 2008, <https://www.wired.com/2008/06/pb-theory/> (accessed 3 May 2026).

²⁰ And the bigger datasets get, the more those biases are amplified, see recent studies like Abeba Birhane, et al., “On Hate Scaling Laws for Data-Swamps,” *arXiv* (2023), arXiv:2306.13141, <https://arxiv.org/abs/2306.13141>.

²¹ Kenneth D. Aiello and Michael Simeone, “Triangulation of History Using Textual Data,” *Isis* 110, no. 3 (2019): 522–37; Julia Damerow and Dirk Wintergrün, “The Hitchhiker’s Guide to Data in the History of Science,” *Isis* 110, no. 3 (2019): 513–21; Gibson, Laubichler, and Maienschein, “Introduction (Focus: Computational History and Philosophy of Science)”; Abraham Gibson and Cindy Ermus, “The History of Science and the Science of History: Computational Methods, Algorithms, and the Future of the Field,” *Isis* 110, no. 3 (2019): 555–66; Manfred D. Laubichler, Jane Maienschein, and Jürgen Renn, “Computational History of Knowledge: Challenges and Opportunities,” *Isis* 110, no. 3 (2019): 502–12; Deryc T. Painter, Bryan C. Daniels, and Jürgen Jost, “Network Analysis for the Digital Humanities: Principles, Problems, Extensions,” *Isis* 110, no. 3 (2019): 538–54.

²² Aiello and Simeone, “Triangulation of History Using Textual Data”; Gibson, Laubichler, and Maienschein, “Introduction (Focus: Computational History and Philosophy of Science)”; Gibson and Ermus, “The History of Science and the Science of History”; Laubichler, Maienschein, and Renn, “Computational History of Knowledge”; Painter, Daniels, and Jost, “Network Analysis for the Digital Humanities.”

²³ Laubichler, Maienschein, and Renn, “Computational Perspectives in the History of Science”; Fotis Jannidis, “Digitale Geisteswissenschaften: Offene Fragen—Schöne Aussichten,” *ZMK Zeitschrift für Medien- und Kulturforschung* 10 (2019): 63–70.

²⁴ Guillermo Restrepo, “Computational History of Chemistry – Part 1: Challenges and Opportunities,” *Ambix*, this issue.

This is why I believe that raising digital and computational literacy within the history of science and knowledge is essential. Such literacy does not require every historian to become a technical specialist. It requires familiarity with the conceptual distinctions between digital and computational methods, the plurality of available approaches, and the assumptions that shape datasets, algorithms, and infrastructures. It also requires engagement with critical digital humanities, critical archival studies, and related fields, where questions of bias, representation, and corpus construction have long been central.²⁵

The same applies in particular to AI. Such tools are socio-technical systems shaped by datasets, funding structures, institutions, and ultimately, their intended purposes of use.²⁶ Issues such as the fact that many algorithms and models are derived from surveillance, military applications and practices of data extractivism, which have involved exploitative labour from data workers in data annotation and content moderation, form part of their history and cannot be separated entirely from their scholarly use.²⁷ Yet, I argue that computational methods should not be reduced to the commercial practices of the technology industry.²⁸ Many

²⁵ For examples see note 11. Many of these interventions have called for better documentation practices for datasets, see Gebru et al., “Datasheets for Datasets”; Lang, “Documenting Datasets as a Tool for Change.”

²⁶ Classical references on critical approaches to AI are, for example, Cathy O’Neil, *Weapons of Math Destruction: How Big Data Increases Inequality and Threatens Democracy* (New York: Crown Publishers, 2016); Virginia Eubanks, *Automating Inequality: How High-Tech Tools Profile, Police, and Punish the Poor* (New York: St. Martin’s Press, 2018); Safiya Umoja Noble, *Algorithms of Oppression: How Search Engines Reinforce Racism* (New York: New York University Press, 2018); For more information on the public hype surrounding recent AI technologies, see Arvind Narayanan and Sayash Kapoor, *AI Snake Oil: What Artificial Intelligence Can Do, What It Can’t, and How to Tell the Difference* (Princeton, NJ: Princeton University Press, 2024); Emily M. Bender and Alex Hanna, *The AI Con: How to Fight Big Tech’s Hype and Create the Future We Want* (New York: Harper, 2025). On the impacts and socio-technical situatedness of AI see Kate Crawford, *Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence* (New Haven: Yale University Press, 2021). On the corporate entanglement of recent technological developments see Karen Hao, *Empire of AI: Dreams and Nightmares in Sam Altman’s OpenAI* (London: Penguin, 2025).

²⁷ Pratyusha Ria Kalluri, William Agnew, Myra Cheng, Kentrell Owens, Luca Soldaini, and Abeba Birhane, “Computer-Vision Research Powers Surveillance Technology,” *Nature* 643 (2025): 73–79, <https://doi.org/10.1038/s41586-025-08972-6>; Mary L. Gray and Siddharth Suri, *Ghost Work: How to Stop Silicon Valley from Building a New Global Underclass* (Boston, MA: Eamon Dolan Books, 2019). On data work in DH see Rafael C. Alvarado, “Datawork and the Future of Digital Humanities,” in *The Bloomsbury Handbook to the Digital Humanities*, ed. James O’Sullivan (London: Bloomsbury Academic, 2022), 361–72, <https://doi.org/10.5040/9781350232143.ch-34>.

²⁸ Pieter Verdegem, “Dismantling AI Capitalism: The Commons as an Alternative to the Power Concentration of Big Tech,” *AI & Society* 39 (2024): 727–37, <https://doi.org/10.1007/s00146-022-01437-8>. While problematic assumptions and biases may be embedded within models themselves, some of the most significant harms associated with contemporary technologies arise primarily from the ways in which they are developed, deployed, and maintained. This includes, for example, exploitative and neo-colonialist forms of data labour, unequal distributions of economic benefit across the Global North and the Global South, and the extractivism in their use of natural resources. Such practices are not necessarily inherent to the underlying technical architecture and could, at least in principle, be organised differently. In many of the most severe cases of harm, it is therefore not the technology itself that is solely responsible but rather the socio-technical systems in which it is situated. Humanities scholars, and those affiliated with or identifying with the humanities more broadly, should therefore make stronger efforts to contribute their expertise to developing of alternative futures for these technologies, as been argued, for example, by Matteo Valleriani, and authors advocating “building otherwise”. See Matteo Valleriani, “Large Language Models That Power AI Should Be Publicly Owned,” *The Guardian*, 26 May 2025, <https://www.theguardian.com/technology/2025/may/26/large-language-models-that-power-ai-should-be-publicly-owned> (accessed 15 June 2026). On “building otherwise,” see Julia Flanders, “Building Otherwise,” in *Bodies of Information: Intersectional Feminism and the Digital Humanities*, ed. Elizabeth Losh and Jacqueline Wernimont, *Debates in the Digital Humanities* (Minneapolis: University of Minnesota Press, 2018), 289–304, <https://doi.org/10.5749/j.ctv9hj9r9.19>. Flanders’ vision has been

digital humanists have been among the most important critics of platform and surveillance capitalism and irresponsible uses of AI.²⁹ They should be seen as allies in developing more reflective, open, and responsible forms of computational scholarship.

In light of this, the challenge ahead is not simply to devise more sophisticated methods. It is also about connecting dispersed communities, improving digital literacy, building reusable datasets, and strengthening the infrastructures on which computational research depends. Earlier programmatic interventions helped create the intellectual space for such a field. The task now is to move beyond aspirations and develop the digital and computational history of science and knowledge as a coherent scholarly enterprise.

Methods

I argue that a digital and computational history of science and knowledge, or of alchemy and chemistry, should not be defined by any single method, scale, or technical paradigm. Indeed, it has been argued that computational history should not be equated only with large-scale, data-driven research.³⁰ Using techniques such as visualisation, dimensionality reduction, statistical modelling, text mining, or network analysis, computational methods can certainly help scholars to analyse bodies of evidence too large for manual inspection;³¹ yet they can also support far more focused forms of inquiry. Such methods may help recover damaged manuscripts, reconstruct fragmentary texts, identify patterns invisible to traditional reading, or generate additional evidence from individual sources. In this way, computational methods can resemble experimental techniques or scientific instruments: they make particular features of a source newly visible.

²⁸ *Continued*

implemented in frameworks such as “Full-Stack Feminism”, see Sharon Webb, “Defining Full Stack Feminism,” *Full Stack Feminism* (Sussex, 2023), <https://doi.org/10.21428/6094d7d2.b65d448a> (accessed 15 June 2026). At the same time, some concerns are more deeply embedded in the technologies themselves. Practices of classification, categorisation, and modelling, for example, raise important epistemological and political questions that cannot be reduced to how they are deployed alone, such as the fact that modelling gender digitally often forces one to resort to binary classifications, see Lang and Suárez Cronauer, “Dataset Audits for Mitigating Data Gaps”; Lang and Suárez Cronauer, “Beyond Data Feminism.” Nevertheless, it is important to recognise that not all technological harms are inherently tied to the technology itself.

²⁹ The broad adoption of AI across society and academia has also been met by criticism, for example Olivia Guest et al., “Against the Uncritical Adoption of ‘AI’ Technologies in Academia,” Zenodo (5 September 2025), <https://doi.org/10.5281/zenodo.17065099>; Sarah Lang, “Critical Concerns for Using LLMs in the (Computational) Humanities and Beyond,” in *Large Language Models for the History, Philosophy, and Sociology of Science: Reflections from a Field in Motion*, ed. Arno Simons et al. (Bielefeld: transcript, 2026).

³⁰ The call in *The History Manifesto* to focus primarily on macrohistory and *longue durée* perspectives was met with considerable criticism, see Jo Guldi and David Armitage, *The History Manifesto* (Cambridge: Cambridge University Press, 2014), <https://doi.org/10.1017/9781139923880>. Fils, Steinová, and Wouters have pointed out that digital methods can also be used to recover and foreground minute details, for example by rendering legible scrolls and other documents that are too damaged to be read by conventional means, see Ivan Flis, Evina Steinová, and Paul Wouters, “Digital Humanities Are a Two-Way Street,” *Isis* 107, no. 2 (2016): 346–48. <https://www.journals.uchicago.edu/doi/pdf/10.1086/687221>.

³¹ For example Painter, Daniels, and Jost, “Network Analysis for the Digital Humanities.” Ruth Ahnert, et al., *The Network Turn: Changing Perspectives in the Humanities* (Cambridge: Cambridge University Press, 2020), <https://doi.org/10.1017/9781108866804>.

Stylometry, a technique for authorship attribution or the study of linguistic style through statistical patterns going back to the 1890s, offers one example: By analysing features such as the distribution of function words, it can produce what are sometimes described as authorial “fingerprints.”³² These patterns are difficult for human readers to detect systematically, but they may provide evidence about authorship and textual relationships, which can support scholarly intuitions concerning authorship by identifying stylistic similarities and differences that would otherwise remain hard to formalise. Such methods do not replace the connoisseurship characteristic of much traditional historical scholarship; rather, they provide another evidential pathway through which expert judgement can be supported, challenged, or refined. The contributions to this special issue therefore illustrate methodological pluralism in practice.

The point raised above equally applies to the studies of Ruland’s *Lexicon Alchemiae* in this special issue. One article examines the structure of the dictionary through a digital and analogue perspective, taking Arabic-derived terminology as a case study and exploring the use of large language models to support the analysis.³³ A companion article approaches the same TEI-XML dataset from another angle, focusing on Paracelsian vocabulary through methods from the distant reading paradigm of quantitative text analysis.³⁴ The contrast is instructive. One approach remains closer to established digital practices, using relatively simple statistical measures and AI-assisted retrieval to support qualitative interpretation. The other uses word embeddings and more explicitly computational methods, producing more abstract results that require a different kind of methodological interpretation. This juxtaposition clearly indicates that there is no hierarchy in which more computational methods are necessarily more valuable. Many digital methods are intuitive and accessible because they extend familiar humanities practices, such as concordancing, word-counting, and classification. Therefore, these methods ultimately support traditional close reading through the interplay of quantitative exploration and qualitative interpretation in the form of hybrid reading.³⁵ Their

³² Sarah Lang, “Assessing Michael Maier’s Contributions to Francis Anthony’s *Apologia* (1616) Using Stylometry,” in *Computational Humanities Research 2021: Conference Proceedings* (Aachen: CEUR-WS, 2021), 346–358, https://ceur-ws.org/Vol-2989/short_paper44.pdf (accessed 3 May 2026).

³³ Sarah Lang, Farzad Mahootian, and Hazem Lashen, “Mediating Alchemical Language across Terminologies and Cultures in Ruland’s *Lexicon Alchemiae*: A Data-Driven Study of Arabic Terms,” *Ambix*; this issue.

³⁴ TEI-XML stands for XML (eXtensible Markup Language) data formatted according to the *Text Encoding Initiative* (TEI) standard, a best practice for encoding texts in the Digital Humanities. Vojtěch Kaše and Sarah Lang, “Contextual Word Embeddings for Paracelsian Lexicography: Tangled Terminologies and their Origins in Ruland’s *Alchemical Dictionary*,” *Ambix*; this issue. Antonija Primorac, Rosario Arias, Roxana Patras, Eva Eglāja-Kristsons, Karina van Dalen-Oskam, Berenike Herrmann, Christof Schöch, and Pieter François, “Distant Reading Two Decades On: Reflections on the Digital Turn in the Study of Literature,” *Digital Studies/Le champ numérique* 13, no. 1 (2023): 1–24, <https://doi.org/10.16995/dscn.8855>. Classical books on distant reading are: Franco Moretti, *Distant Reading* (Verso Books, 2013). Matthew L. Jockers, *Macroanalysis: Digital Methods and Literary History* (Champaign: University of Illinois Press, 2013). Ted Underwood, *Distant Horizons: Digital Evidence and Literary Change* (Chicago: University of Chicago Press, 2019). On close reading, see Karine Chemla, ed., *History of Science, History of Text* (Springer Netherlands, 2005), <https://doi.org/10.1007/1-4020-2321-9>.

³⁵ On hybrid methodologies in Distant Reading see Parham Aledavood, “Taking the Middle Road: Reflections on Mixed Methodology within the Digital Humanities,” *Digital Studies/Le champ numérique* 14, no. 1 (2024): 1–19, <https://doi.org/10.16995/dscn.11069> and Georgiana D. Hedesan, “Fire, Vulcanus, Archeus, and Alchemy: A

value lies not in making possible what could never be done manually, but in accelerating exploratory work, structuring evidence, and creating datasets or subsets for further analysis. A workflow with a “human in the loop,” such as the controlled use of large language models followed by human verification, can free time for the interpretative work that remains central to humanities research.³⁶ For this reason, I hold that computational methods should not be understood solely as tools for analysing pre-existing datasets. In many digital humanities contexts, they are also being used to enrich data, impose structure on previously unstructured materials, support annotation and classification, or generate entirely new forms of evidence.³⁷

It is useful, here, to distinguish between exploratory and more controlled uses of computation.³⁸ Some methods function heuristically: they help researchers locate material, identify patterns, and generate questions. Others are closer to experiments, testing specific hypotheses under more clearly defined conditions. Neither mode is inherently superior. Heuristic digital methods often produce enriched datasets or structured collections of evidence, while more computationally intensive approaches may generate measurements or abstract outputs that themselves require

³⁵ *Continued*

Hybrid Close-Distant Reading of Paracelsus’s Thought on Active Agents,” *Ambix* 71, no. 3 (2024): 271–300, <https://doi.org/10.1080/00026980.2024.2367396>.

³⁶ Eduardo Mosqueira-Rey, et al., “Human-in-the-Loop Machine Learning: A State of the Art,” *Artificial Intelligence Review* 56, no. 4 (2023): 3005–54, <https://doi.org/10.1007/s10462-022-10246-w>. Of course, computational methods must never be treated as objective or assumption-free. As with chemical analysis, one only obtains evidence about the properties for which one chooses to test. Datasets, parameters, models, and categories all shape the results. Historical data are never raw: they have passed through stages of selection, preservation, digitisation, transcription, encoding, and interpretation. Computational analysis may reveal patterns within such data, but it cannot escape the conditions under which the data were produced. Like experimental research in the history of science, computational inquiry is fundamentally iterative. Methods are rarely applied once to a fixed body of material and then allowed to produce results automatically. Instead, computational workflows typically involve repeated cycles of data preparation, analysis, evaluation, and revision. Data preparation is often the most labour-intensive stage of the process, and researchers frequently return to the data after initial analyses in order to refine, correct, or augment them.

³⁷ See for example Sarah Lang, “Towards a Data-Driven History of Lexicography: Two Alchemical Dictionaries in TEI-XML,” *Journal of Open Humanities Data* 11, no. 20 (2025): 1–6, <https://doi.org/10.5334/johd.303>; Sarah Lang, “Fine-Tuning Machine Learning with Historical Data: An Alchemical Object Detection Dataset for Early Modern Scientific Illustrations,” *Zeitschrift für digitale Geisteswissenschaften* 10 (2025), https://doi.org/10.17175/2025_002. Vojtěch Kaše, Sarah Lang, and Petr Pavlas, “Embedded in the Labyrinth: Investigating Latin Word Senses through Transformer-Based Contextual Embeddings and Attention,” in *Proceedings of the Computational Humanities Research Conference 2025*, 498–512, <https://doi.org/10.63744/FuaAvdPMdtwW>; Lang et al., “Confabulated Transliterations?”

³⁸ As has been suggested for experimental methods in the history of science, it is useful to distinguish between *experiencing* and *experimenting*. Applied to computational approaches, exploratory and heuristic methods are better understood as forms of *experiencing*, whereas computational methods can also be deployed in more targeted and controlled ways that more closely resemble *experimenting*. In such cases, methods are used to test specific hypotheses under carefully defined conditions. On the difference between *experiencing* and *experimenting* in reconstruction, replication, and re-enactment (RRR) methods, see: Hjalmar Fors, Lawrence M. Principe, and H. Otto Sibum, “From the Library to the Laboratory and Back Again: Experiment as a Tool for Historians of Science,” *Ambix* 63, no. 2 (2016): 85–97 (on 93), <https://doi.org/10.1080/00026980.2016.1213009>; Marieke M. A. Hendriksen, “Rethinking Performative Methods in the History of Science,” *Berichte zur Wissenschaftsgeschichte* 43, no. 3 (2020): 313–22 (on 317), <https://doi.org/10.1002/bewi.202000017>. For the application of the concept to digital experiments, see Sarah Lang, “Experiments in the Digital Laboratory: What the Computational Humanities Can Learn about Their Definition and Terminology from the History of Science,” in *Fabrikation von Erkenntnis: Experimente in den Digital Humanities*, ed. Manuel Burghardt, et al. (Luxembourg: Melusina Press, 2022), <https://doi.org/10.26298/vra6-mn32>.

interpretation. Both require rigour, and both depend on the fit between method, source, and research question.

Work by Guillermo Restrepo, for instance, ranges from semantic tracking of terms to citation network analysis, combining text mining with bibliometric approaches.³⁹ The EMLAP corpus, discussed elsewhere in this special issue, foregrounds the scholarly labour of dataset creation.⁴⁰ The preparation of structured data emerges as a central step that will allow future scholars to run computational methods on a representative sample of sixteenth century alchemical print. Rather than being purely technical, the data preparation closely resembles traditional scholarly work and constitutes a substantial part of the research effort. Jeffrey Johnson's contribution situates quantitative methods in the history of chemistry within a longer historiographical background.⁴¹ Together, these studies show that the future of the field lies not in identifying a single preferred algorithm or workflow, but in building the conditions under which different methods can be brought into productive dialogue.

One of the distinctive possibilities of computational history of science and knowledge is that it can draw not only on methods developed within digital humanities, but also on methods from the sciences themselves.⁴²

Data and diversity

A central challenge for a digital and computational history of science and knowledge concerns questions of temporal, linguistic, and cultural coverage. Much existing computational work relies on relatively recent material, typically from the past one or two centuries.⁴³ This is not accidental. It is well-known that many

³⁹ Guillermo Restrepo, "ComputationalHistory of Chemistry – Part 1"; Guillermo Restrepo, "Computational History of Chemistry – Part 2: Multiple Scales, from the Individual to the Global," *Ambix*, this issue. Similar directions are explored by Adrian Wüthrich, pointing to the potential of such methods while also illustrating their current areas of application. Arno Simons, Michael Zichert, and Adrian Wüthrich, "Large Language Models for History, Philosophy, and Sociology of Science: Interpretive Uses, Methodological Challenges, and Critical Perspectives," *Studies in History and Philosophy of Science* 117 (2026), <https://doi.org/10.1016/j.shpsa.2026.102151>; Arno Simons, et al., eds., *Large Language Models for the History, Philosophy, and Sociology of Science: Reflections from a Field in Motion* (Bielefeld: transcript, 2026), <https://www.transcript-verlag.de/978-3-8376-7994-6/understanding-science-with-large-language-models/?c=313000027> (accessed 8 June 2026).

⁴⁰ Georgiana Hedesan, Vojtěch Kaše and Petr Pavlas, "An Alchemical Prima Materia for the Digital Age: Making the EMLAP (Early Modern Latin Alchemical Prints) Dataset," *Ambix*, this issue.

⁴¹ Jeffrey A. Johnson, "Emergence of New Quantitative Approaches to the History of Chemistry after 1950," *Ambix*, this issue. For the history of quantitative history, see for example: William O. Aydelotte, "Quantification in History," *American Historical Review* 71, no. 3 (1966): 803–25; François Furet, "Quantitative History," *Daedalus* 100, no. 1 (1971): 151–67; Steven Ruggles, "The Revival of Quantification: Reflections on Old New Histories," *Social Science History* 45, no. 1 (2021): 1–25, <https://doi.org/10.1017/ssh.2020.44>; Relatedly also Daniel Allington, "The Place of Computation in the Study of Culture," in *The Bloomsbury Handbook to the Digital Humanities*, ed. James O'Sullivan (London: Bloomsbury Academic, 2022), 373–84.

⁴² This potential was already noted in the 2019 *Isis* focus section discussed earlier, where evolutionary biology was presented as a useful source of concepts and models for studying change over time. Yet the same point applies beyond the life sciences. In the history of alchemy and chemistry, for example, computational chemistry methods may be repurposed for historical inquiry, as co-editors of this special issue Guillermo Restrepo and Sarah Lang explore in forthcoming work for a special issue of *Notes and Records* on computational and experimental methods in the history of science. Such work suggests that the field can generate new forms of exchange between scientific methods and humanities scholarship.

computational methods, including those associated with AI, perform best on modern, standardised, and well-structured data in well-supported languages such as English.⁴⁴ As a result, computational research is often shaped by the datasets and tools that happen to be available rather than by the full diversity of historical questions that scholars might ideally like to ask.

These biases are not only temporal but also linguistic and cultural. Computational approaches tend to work best on English-language materials and on source traditions that already benefit from extensive digital infrastructures. Pre-modern materials, less well-resourced languages, and many non-Western knowledge traditions remain comparatively difficult to analyse computationally.⁴⁵ Consequently, scholars working on such sources do not operate on equal terms with those studying modern scientific corpora, and existing disparities are reinforced by the uneven distribution of datasets and digitisation efforts.⁴⁶ Computational methods depend on the availability of structured datasets, which in turn reflect existing priorities in archival record-keeping and funding.⁴⁷ Similar challenges apply to sources beyond text, where the necessary infrastructures are often even less developed.⁴⁸ The result is an uneven landscape in which some areas of the history of science and knowledge are significantly more amenable to computational investigation than others.

The implications are visible in the kinds of questions computational approaches have most readily addressed. Much existing work has concentrated on topics such

⁴³ Taylor Arnold and Lauren Tilton, "Distant Viewing: Analyzing Large Visual Corpora," *Digital Scholarship in the Humanities* 34 (2019): 3–16; Melvin Wevers and Thomas Smits, "The Visual Digital Turn: Using Neural Networks to Study Historical Images," *Digital Scholarship in the Humanities* 35 (2020): 194–207; Thomas Smits and Melvin Wevers, "A Multimodal Turn in Digital Humanities: Using Contrastive Machine Learning Models to Explore, Enrich, and Analyze Digital Visual Historical Collections," *Digital Scholarship in the Humanities* 38 (2023): 1267–80; Taylor Arnold and Lauren Tilton, *Distant Viewing: Computational Exploration of Digital Images* (Cambridge, MA: MIT Press, 2023). For an overview contextualizing how the relevant discourses are related see: Lang, "(Doing) Computational History."

⁴⁴ Domenico Fiormonte, "Taxation against Overrepresentation? The Consequences of Monolingualism for Digital Humanities," in *Alternative Historiographies of the Digital Humanities*, ed. Dorothy Kim and Adeline Koh (Santa Barbara: Punctum Books, 2021), 333–376, <https://www.jstor.org/stable/j.ctv1r7878x.13>.

⁴⁵ For example, digital work on the early modern German of the so-called *Processus Universalis* alchemical recipe group, studied by the Gotha Alchemy Network as part of a 2018–2019 project, was initially severely hampered by the non-standardised orthography of German during the period under investigation. Today, AI-powered algorithms can address many of these challenges far more effectively. However, multilingual and non-Western materials still frequently suffer malfunctioning algorithms. See Sarah Lang, "Zwischen Experimental und Computational History of Science: Digitales Experimentieren an alchemischen Prozessvorschriften," in *Veröffentlichungen des Landesamtes für Denkmalpflege und Archäologie Sachsen-Anhalt*, vol. 101 (2026): 69–85; Lang et al., "Confabulated Transliterations?"

⁴⁶ On missingness in data, see Alexandra Ortolja-Baird and Julianne Nyhan, "Encoding the Haunting of an Object Catalogue: On the Potential of Digital Technologies to Perpetuate or Subvert the Silence and Bias of the Early-Modern Archive," *Digital Scholarship in the Humanities* 37, no. 3 (2022): 844–67; Rebecca Sutton Koeser and Zoe LeBlanc, "Missing Data, Speculative Reading," *Journal of Cultural Analytics* 9, no. 2 (2024), <https://doi.org/10.22148/001c.116926>; Mike Kestemont, et al., "Forgotten Books: The Application of Unseen Species Models to the Survival of Culture," *Science* 375, no. 6582 (2022): 765–69, <https://doi.org/10.1126/science.abl7655>.

⁴⁷ Gerben Zaagsma, "Digital History and the Politics of Digitization," *Digital Scholarship in the Humanities* 38 (2023): 830–51; Lang and Suárez Cronauer, "Dataset Audits for Mitigating Data Gaps."

⁴⁸ Ruta Binkytė, "Distant Reading and Viewing: Big Questions in Digital Art History and Digital Literary Studies," *Digital Humanities Quarterly* 17, no. 3 (2023): 1–5.

as interdisciplinarity, scientific collaboration, innovation, and citation networks within modern scientific communities.⁴⁹ Such questions are important, but they are also closely tied to the availability of bibliometric data and to forms of evidence that privilege modern scientific institutions. Consequently, the computational history of science has often overlapped with scientometrics and bibliometrics.⁵⁰ While valuable, this evidential base remains relatively narrow when compared with the broader ambitions of a global history of science and knowledge.⁵¹

If the field is to contribute meaningfully to a more inclusive history of knowledge, these limitations must be addressed directly. Citation networks and related methods, for example, often reproduce established narratives centred on recognised innovators and highly visible actors. Although they can occasionally help recover overlooked figures, they do not automatically challenge deeper structural biases in the historical record. Doing so frequently requires more fundamental interventions, including the digitisation, annotation, and preparation of sources that have not previously been prioritised.⁵²

While these aforementioned types of research scenarios are presented as broadly relevant to the history of science, they are closely tied to modern research environments, citation practices, the availability of bibliometric data and lend themselves particularly well to writing hero narratives of lone scholars that many in the history of knowledge would like to move beyond.⁵³ There is also a tension

⁴⁹ Examples can be found in: Restrepo, “Computational History of Chemistry – Part 1”; Restrepo, “Computational History of Chemistry – Part 2”; Simons, Zichert, and Wüthrich, “Large Language Models for History, Philosophy, and Sociology of Science”; Simons et al, *Large Language Models for the History, Philosophy, and Sociology of Science*.

⁵⁰ On scientometrics and bibliometrics see William W. Hood and Concepción S. Wilson, “The Literature of Bibliometrics, Scientometrics, and Informetrics,” *Scientometrics* 52, no. 2 (2001): 291–314.

⁵¹ Lorraine Daston, “The History of Science and the History of Knowledge,” *KNOW: A Journal on the Formation of Knowledge* 1, no. 1 (2017): 131–54, <https://doi.org/10.1086/691678>; On a more global history of alchemy, see also recently Alexandre M. Roberts, “Disentangling Alchemy,” *Isis* 116, no. 4 (2025): 623–43, <https://doi.org/10.1086/738213>.

⁵² AI ethics approaches that focus primarily on explanation (such as explainable AI) have frequently been criticised for merely diagnosing problems without addressing their underlying causes. A growing body of scholarship has argued that many of these issues originate in the datasets upon which computational methods are built, suggesting that meaningful improvements require greater attention to data quality, data practices, and the conditions under which data are collected, curated, and used. For an example of explainable AI, see Hassan El-Hajj et al., “Explainability and Transparency in the Realm of Digital Humanities: Toward a Historian XAI,” *International Journal of Digital Humanities* 5 (2023): 299–331, <https://doi.org/10.1007/s42803-023-00070-1>. For the aforementioned criticisms see: Catherine D’Ignazio and Lauren F. Klein, *Data Feminism* (Cambridge, MA: MIT Press, 2020); Emily M. Bender, et al., “On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?,” in *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT ’21)*, ed. Souneil Park and Min Lee (New York: Association for Computing Machinery, 2021), 610–23; Amandalynne Paullada, et al., “Data and Its (Dis)Contents: A Survey of Dataset Development and Use in Machine Learning Research,” *Patterns* 2 (2021): 100336; Lauren F. Klein and Catherine D’Ignazio, “Data Feminism for AI,” in *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency (FAccT ’24)*, ed. Reuben Binns et al., (New York: Association for Computing Machinery, 2024), 100–12; Sarah Lang, “Critical Concerns for Using LLMs in the (Computational) Humanities and Beyond,” in *Large Language Models for the History, Philosophy, and Sociology of Science: Reflections from a Field in Motion*, ed. Arno Simons et al. (Bielefeld: transcript, 2026), 35–50, <https://www.transcript-verlag.de/978-3-8376-7994-6/understanding-science-with-large-language-models/?c=313000027> (accessed 8 June 2026).

⁵³ For instance, they may render invisible the contributions of individuals involved in research projects who are not granted authorship, a practice that has long been normalised within academic scholarship. Only recently has this begun to change through initiatives such as the CRediT taxonomy, which enables the creation of more nuanced

between the specificity of computational analyses and the generality with which their findings are sometimes presented. Results are always shaped by particular datasets, algorithms, and modelling choices. Yet computational approaches are occasionally framed as offering broader insights into the history of science as a whole. Greater precision is therefore needed when situating such work. Computational methods do not constitute a single, uniform methodology but rather a diverse collection of approaches whose applicability varies considerably depending on the materials under investigation.

This returns us to the importance of methodological diversity to reflect the diversity of sources and research questions we would like to deal with. The contributions to this special issue illustrate that broadening the evidential base often begins with the creation of new datasets rather than with the application of increasingly sophisticated algorithms. The articles on Ruland's *Lexicon Alchemiae* focusing on Arabic terminology and multilingual knowledge transfer, as well as the contribution describing the rationale behind the creation of the EMLAP dataset in this special issue, highlight the extent to which computational analysis depends upon prior scholarly labour.⁵⁴ The creation of usable datasets remains a prerequisite for many advanced forms of analysis.

Recognising these conditions also means resisting the assumption that more technically complex methods are inherently superior. Such an assumption risks reinforcing existing inequalities because underrepresented, multilingual, and non-Western sources often require substantial preparatory work before they can be analysed computationally. A more balanced field therefore depends upon deliberate choices regarding which sources are digitised, curated, and transformed into reusable datasets. In many cases, materials that are currently least amenable to computation may offer the greatest potential for expanding the scope of the field.

The challenge, then, is not merely methodological but also epistemological. A digital and computational history of science and knowledge must remain attentive to the ways in which data availability shapes historical inquiry. Addressing biases in datasets, infrastructures, and research design is essential if the field is to move beyond its current concentration on modern, well-documented contexts and towards a genuinely broader, more global, and more inclusive history of knowledge.

⁵³ *Continued*

contributor profiles and allows certain forms of labour and responsibility to be formally acknowledged and documented that were not credited before. See also: Julianne Nyhan, "On the Making of the Myth of the Lone Scholar: Digital Humanities as Aetiology," in *Hidden and Devalued Feminized Labour in the Digital Humanities: On the Index Thomisticus Project, 1954–67* (London: Routledge, 2022): <https://www.taylorfrancis.com/chapters/mono/10.4324/9781003138235-6/>. Approaching this topic following the "Invisible Technician" discourse, see Tessa Gengenagel and Sarah Lang, "A Discipline, Divided: On the Digital Humanities and Ideologies of Knowledge Work," in *DH2026 Book of Abstracts* (2026): 486–489; Sarah Lang, "Two-tier Computational Humanities: A Labour History of Undervalued Contributions in DH," in *Handbook of Feminist Digital Scholarship*, ed. Anne Cong-Huyen and Kim Knight (Berlin: De Gruyter, forthcoming 2027), drawing on Steven Shapin, "The Invisible Technician," *American Scientist* 77, no. 6 (1989): 554–63.

⁵⁴ Lang, Mahootian, and Lashen, "Mediating Alchemical Language across Terminologies and Cultures"; Kaše and Lang, "Contextual Word Embeddings for Paracelsian Lexicography."

Closing remarks

The central argument of this epilogue has been that, despite a growing body of scholarship and a number of influential programmatic interventions, a fully established digital and computational history of science and knowledge does not yet exist. What exists instead are many of the components from which such a field could be built: datasets, projects, methods, infrastructures, and communities of expertise. I have argued that the task ahead is to bring these elements into closer dialogue and to develop a more coherent intellectual and institutional framework for their interaction.

One priority is *conceptual clarity*. Discussions of computational history of science have often focused primarily on computational methods, yet computational approaches represent only one part of a broader digital ecosystem. Computational humanities depends upon digital humanities just as computational analysis depends upon datasets, metadata, editions, and infrastructures. For this reason, the future of the field lies not in a computational history of science alone, but in a digital and computational history of science, knowledge, and technology that recognises the interdependence of data creation and data analysis.

A second priority is *diversity*: of methods, sources, languages, periods, and forms of knowledge. Computational approaches will only fulfil their broader potential if they move beyond the relatively narrow range of materials currently best supported by existing tools and infrastructures. This requires continued investment in dataset creation, curation, and documentation, particularly for underrepresented languages, regions, and knowledge traditions. The calls for a shift from model-centric to data-centric thinking in contemporary AI underscores the importance of this task.⁵⁵ Better algorithms cannot truly compensate for limitations in the underlying evidence.

The development of *shared infrastructures* will be equally important. Projects such as Transkribus, which was used to create the EMLAP dataset presented in this special issue, demonstrate that scholarly communities can collaboratively create training data, datasets, and tools that serve collective research needs.⁵⁶ Such initiatives also illustrate that activities often considered peripheral to computational research like transcription, annotation, editing, and metadata creation are in fact fundamental scholarly contributions. Computational methods depend upon this work, just as much as they depend upon advances in modelling and analysis.

For this reason, computational approaches should not be viewed as occupying the summit of a methodological hierarchy. They depend upon the broader

⁵⁵ Mohammad Hossein Jarrahi, Ali Memariani, and Shion Guha, “The Principles of Data-Centric AI (DCAI),” *Communications of the ACM* 66 (2023): 84–92. Johannes Jakubik, Michael Vössing, Niklas Köhl, Jannis Walk, and Gerhard Satzger, “Data-Centric Artificial Intelligence,” *Business & Information Systems Engineering* 66 (2024): 507–15.

⁵⁶ Whatever concerns may exist regarding governance, sustainability, or commercialisation, such projects show that digital infrastructures can be built through collaboration and that the labour involved in creating high-quality data is itself a form of scholarly contribution.

ecosystem of digital scholarship from which their data and infrastructures emerge. Computational analysis relies upon the foundations created through digitisation, editing, transcription, annotation, metadata creation, and dataset construction. These activities are complementary rather than competing. A mature digital and computational history of science and knowledge will require all of them. Dataset builders, editors, archivists, digital scholars, and computational researchers each contribute to the same broader scholarly ecosystem.

At the same time, the future of the field lies both in adopting methods from digital humanities and in drawing creatively on methods developed within the sciences themselves. Historians of science occupy a distinctive position in this regard.⁵⁷ Yet, the field can do more than borrow methods from neighbouring disciplines; it can also create new forms of exchange between scientific and historical modes of inquiry.

Many of these themes are already visible in the contributions to this special issue. The volume broadens the empirical foundations of computational history of science by extending discussion beyond the life sciences and into the histories of alchemy and chemistry. It also demonstrates considerable methodological diversity. Some contributions focus on dataset construction and corpus design, others on textual and network analysis, and still others on connecting computational approaches with the material dimensions of chemical knowledge. Taken together, they illustrate that there is no single computational method, only different combinations of methods suited to different historical problems.

There are, moreover, reasons for cautious optimism. Historical sources will always remain incomplete, and no dataset will ever offer comprehensive coverage of the past.⁵⁸ Yet more material is available than at any previous point in history, and many computational tools have become increasingly accessible. The greatest promise of the current AI moment, I argue, is not that machines will replace historians or automate interpretation. Rather, it lies in enabling scholars to combine existing tools into flexible research workflows tailored to particular questions and sources. Increasingly, researchers with modest technical expertise can integrate transcription tools, textual analysis, network methods, visualisation techniques, and other approaches into coherent analytical pipelines.

Perhaps the most important lesson to emerge from both this special issue and the wider literature is that computational approaches are most productive when understood as complements rather than replacements for established forms of historical scholarship. The strongest results arise neither from algorithms

⁵⁷ Earlier discussions already pointed to the potential of concepts and modelling approaches derived from fields such as evolutionary biology, see for example Laubichler, Maienschein, and Renn, "Computational History of Knowledge." New possibilities emerge through the application of scientific computational methods to historical materials. Forthcoming work by co-editors Guillermo Restrepo and Sarah Lang, for example, explores how computational chemistry can be brought into dialogue with the history of alchemy and chemistry.

⁵⁸ Gerben Zaagsma, "Digital History and the Politics of Digitization," *Digital Scholarship in the Humanities* 38, no. 2 (2023): 830–51, <https://doi.org/10.1093/llc/fqaco50>; Toussaint Nothias, "An Intellectual History of Digital Colonialism," *Journal of Communication* 75, no. 5 (2025): 385–97, <https://doi.org/10.1093/joc/fqaf003>.

operating independently nor from historians working without computational assistance, but from collaboration between human expertise and computational methods.⁵⁹ Historians of science and knowledge are particularly well placed to engage in such collaborations because the field already possesses long traditions of reflecting critically on instruments, models, techniques, and knowledge-making practices.

Therefore, the challenge is no longer to argue that a digital and computational history of science and knowledge should exist. The challenge is to build the intellectual, social, and infrastructural foundations that will allow it to flourish. Doing so will require stronger connections between digital humanities and history of science, between computational and non-computational scholars, between data creation and data analysis, and between different knowledge traditions, languages, and regions. If these connections can be sustained, the field will be well positioned not only to adopt new methods, but also to contribute new questions, new forms of evidence, and new perspectives on the history of science and knowledge itself.

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Statement on the use of AI

During the preparation of this manuscript, the author has used ChatGPT 4o for proofreading and improving the English language. The author has reviewed and edited the output and take full responsibility for the content of this publication.

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⁵⁹ Yannis Assael, Thea Sommerschild, et al., "Restoring and Attributing Ancient Texts Using Deep Neural Networks," *Nature* 603, no. 7900 (2022): 280–83, <https://doi.org/10.1038/s41586-022-04448-z>.

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